

Optimal broadcasting on incomplete star graph interconnection networks

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Abstract

Broadcasting is an important collective communication operation in many applications which use parallel computing. In this paper, we focus on designing broadcasting algorithms for general incomplete star graphs. We propose two optimal one-to-all broadcasting algorithms for incomplete star graphs with a single-port communication model. An incomplete star graph with N nodes, where $(n-1)! < N < n!$, is a subgraph of an n -dimensional star graph. The first scheme is single-message one-to-all broadcasting that takes $O(n \log n)$ steps. The second one is multi-message one-to-all broadcasting that takes $O(n \log n + m)$ steps.

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1. Introduction

Broadcasting is an important collective communication operation in many applications which use parallel computing. Much research has given attention to the problems associated with achieving

high performance computing in broadcasting and personalized communication on a variety of interconnection networks [4,5,8,20].

The star graph [1,2] interconnection network has been recognized as an attractive alternative to the popular hypercube network. More topological properties of the star graph can be found in [1]. Much effort has been extended to designing broadcasting algorithms for star graph interconnection networks [11,13–15,17,19,20]. Methods of constructing spanning trees in star graphs have also been proposed to solve the problems

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associated with personalized communication [4–6]. In addition, there has also been some research on fault-tolerant routing and fault-tolerant broadcasting [3,7,12,16,21].

Although interconnection networks, including hypercubes, star graphs, and WK-recursive networks have many nice properties, there is a serious restriction on the number of nodes that they can have. To relieve this restriction and eliminate the gap between two consecutive sizes of a given topology, a variety of incomplete interconnection networks, such as incomplete hypercubes [9], incomplete star graphs [10], and incomplete WK-recursive networks [18], all of which can have any number of nodes, have been proposed. In [9], Katseff proposed a broadcasting algorithm for incomplete hypercubes. In [18], Su et al. proposed a broadcasting algorithm for incomplete WK-recursive networks. In [10], Latifi et al. proposed a broadcasting algorithm for a special class of incomplete star graphs.

In this paper, we focus on designing broadcasting algorithms for general incomplete star graphs. We propose two optimal one-to-all broadcasting algorithms for incomplete star graphs with a single-port communication model. An incomplete star graph with N nodes, where $(n-1)! < N < n!$, is a subgraph of an n -dimensional star graph. When an incomplete star graph is partitioned into several substar graphs in order to deliver a message, optimal single-message broadcasting in $O(n \log n)$ steps is addressed. Through this broadcasting scheme, we can generalize a previous work [10] which focused on a special class of incomplete star graphs. We also present an optimal multi-message broadcasting algorithm in $O(n \log n + m)$ steps. With this scheme, m messages can be broadcasted on an incomplete star graph. Multi-message broadcasting is done first by transmitting m messages to each substar in a pipeline fashion and then by using the algorithm proposed in [14] to broadcast them.

The rest of this paper is organized as follows. Section 2 introduces the definitions, notations, and lemmas used in this paper. In Section 3, we propose two optimal one-to-all broadcasting algorithms for incomplete star graphs. Finally, conclusions are given in Section 4.

2. Preliminaries

In this section, we first introduce some definitions and notations related to star graphs and incomplete star graphs. A permutation of n distinct symbols from the set $\{1, 2, \dots, n\}$ is given as $p = p_1 p_2 \dots p_n$, where $p_i, p_j \in \{1, 2, \dots, n\}$, $p_i \neq p_j$ for $i \neq j$, $1 \leq i, j \leq n$. Given a permutation $p = p_1 p_2 \dots p_n$, let the generator g_i be the function of p that interchanges the symbol p_i and the symbol p_1 in p for $2 \leq i \leq n$. Thus, $g_i(p) = p_i p_2 \dots p_{i-1} p_1 p_{i+1} \dots p_n$. An undirected star graph (star) with dimension n is denoted as $S_n = (V_n, E_n)$, where the set of vertices V_n is defined as $\{v | v = p_1 p_2 \dots p_n | p_i, p_j \in \{1, 2, \dots, n\}, p_i \neq p_j \text{ for } i \neq j, 1 \leq i, j \leq n\}$ and the set of edges E_n is defined as $\{(v_p, v_q) | v_p, v_q \in V_n, v_p \neq v_q, \text{ such that } v_q = g_i(v_p) \text{ for } 2 \leq i \leq n\}$.

In other words, any two nodes v_p and v_q are connected by an undirected edge if and only if the permutation of the node v_q can be obtained from the corresponding permutation of v_p by interchanging the symbol p_i of v_p and the symbol p_1 of v_p for $2 \leq i \leq n$. We also use the notation S_n to represent an n -dimensional star graph, called an n -star, in this paper. Note that star graphs are edge and vertex symmetric. Moreover, S_n is a regular graph with a degree of $n-1$, $n!$ vertices, $\frac{(n-1)n!}{2}$ edges, and a diameter of $\lfloor \frac{3(n-1)}{2} \rfloor$.

Let $p_1 p_2 \dots p_{n-i}$ be a permutation of $n-i$ distinct symbols in $\{1, 2, \dots, n\}$ for $1 \leq i \leq n$. The substar graph (substar), denoted by $S_n^i(p_1^{a_1}, p_2^{a_2}, \dots, p_{n-i}^{a_{n-i}})$, is defined as a subgraph (V, E) of S_n . Note that V is the set of nodes with the same $n-i$ symbols p_1 in position a_1 , p_2 in position a_2 , ..., and p_{n-i} in position a_{n-i} , and E is the set of edges incident with any two of those nodes in V . Occasionally, we may denote $S_n^i(p_1^{a_1}, p_2^{a_2}, \dots, p_{n-i}^{a_{n-i}})$ by a sequence $x_1 x_2 \dots x_n$ such that for all $1 \leq i \leq k$, $x_{a_i} = p_i$, and for all $j \notin \{p_1, p_2, \dots, p_k\}$, $x_j = *$, where a $*$ means “don’t care”. For instance, $S_4^2(3^2, 1^4)$ is denoted as $*3*1$.

We define an incomplete star graph (incomplete star) as follows [10]. An important aspect of an incomplete star graph is the methodology for constructing such a graph with an arbitrary number of nodes. Assume that N is the total number of nodes in an incomplete star graph, where

$(n-1)! < N < n!$ for some integer n . In this construction, we define the coefficient vector as an $(n-1)$ -tuple vector in which the k th element of the vector is the number of $(n-k)$ -star, where $1 \leq k \leq n-1$, in the incomplete star graph. We suppose that the coefficient vector is $(n-1)$ -tuple $(b_{n-1}, b_{n-2}, \dots, b_2, b_1)$, where $0 \leq b_i \leq i$, such that $N = b_{n-1}(n-1)! + b_{n-2}(n-2)! + \dots + b_2(2!) + b_1(1!) = \sum_{i=1}^{n-1} b_i(i!)$. That is, b_{n-1} $(n-1)$ -stars, b_{n-2} $(n-2)$ -stars, $\dots$, b_1 1-stars exist in an N -node incomplete star graph.

The labeling scheme for each node in the incomplete star graph has been described in [10]. The stars with the same dimension k are classified to form a class C^k . According to such a labeling scheme, two consecutive labels (stars) are said to be adjacent. Note that two adjacent substars need not belong to the same class. For example, a 17-node incomplete star graph with the coefficient vector $(b_3, b_2, b_1) = (2, 2, 1)$ consists of $C^3 = \{***4, ***3\}$, $C^2 = \{**12, **42\}$, and $C^1 = \{4132\}$, as shown in Fig. 1. The two substars $**12$ and $**42$ are adjacent in the same class C^2 . The two substars $**42$ and 4132 are adjacent in different classes. For simplicity, we use the notation $*^k c_{k+1} \dots c_n$ to denote the k -dimensional substar $*** \dots * c_{k+1} \dots c_n$. Without loss of generality, the function $next()$, as in [10], denotes, in a lexicographical order, the

modulo operation for the unused integers. For the two adjacent substars $**12$ and $**42$ in the above figure, the value of $next(1)$ is 4, that is, $**next(1)2 = **42$.

In the class C^k , we can number the substars $C_1^k, C_2^k, \dots$, and $C_{b_k}^k$ according to the labeling scheme. We call C_i^k the i th substar in C^k . Hence, the two substars C_i^k and C_{i+1}^k , where $1 \leq i < b_k$, are adjacent. We claim that each pair of substars in the same class has the adjacency property. That is, any node in substar C_i^k via constant links can connect to one of the nodes in substar C_j^k , where $i \neq j$.

Lemma 1. Suppose that two substars C_i^k and C_j^k exist in the same class C^k , where $i \neq j$ and $1 \leq i, j \leq b_k$. Routing is done in at most 2 steps from any node in C_i^k to one of the nodes in C_j^k .

Proof. Assume that $C_i^k = *^k a c_{k+2} \dots c_n$ and $C_j^k = *^k b c_{k+2} \dots c_n$. Suppose that a node $c = x_1 x_2 \dots x_k a c_{k+2} \dots c_n$ exists in C_i^k , where $x_s = b$ and $1 \leq s \leq k$. In routing a message, the first step is to route the message from node c to node $d = g_s(c) = b x_2 \dots x_{s-1} x_{s+1} \dots x_k a c_{k+2} \dots c_n$ in C_i^k . Then, the second step is to route the message from node d to node $e = g_{k+1}(d) = a x_2 \dots x_{s-1} x_{s+1} \dots x_k b c_{k+2} \dots c_n$. Clearly, node e belongs to the substar C_j^k . Thus, this routing is done in at most two steps. $\square$

Corollary 1. When a message is simultaneously sent on any two pairs of substars in C^k , no conflicts can occur.

Proof. During routing in each substar in C^k , no conflicts can occur in any first step of the routing. The second step of the routing uses the generator g_{k+1} to send the message, because the second step does not use the same links in C^k , where $1 \leq k \leq n-1$. Thus, the proof of this corollary holds. $\square$

In the following, we also claim that two substars according to the labeling scheme in the two consecutive classes also have the adjacency property. That is, any node in the last substar $C_{b_i}^i$ can connect to one of nodes in the first substar C_1^j via $O(i-j)$ links, where $i > j$ [10].

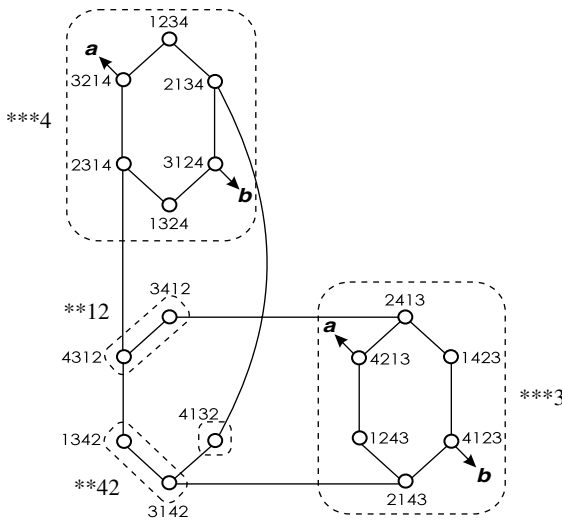


Fig. 1. An incomplete star graph with 17 nodes.

Lemma 2. Suppose that two contiguous classes C^i and C^j exist for $i > j$. Routing is done in at most $2(i - j) + 2$ steps from any node in the last substar $C_{b_i}^i$ to one of the nodes in the first substar C_1^j , according to the labeling scheme.

Proof. Assume that a node s is of the form $s = *^i a_{i+1} a_{i+2} \dots a_n$ in $C_{b_i}^i$ and a node d is of the form $d = *^j x_{j+1} x_{j+2} \dots x_i \text{next}(a_{i+1}) a_{i+2} \dots a_n$ in C_1^j . Routing from s to s' in the form $s' = *^j x_{j+1} x_{j+2} \dots x_i a_{i+1} a_{i+2} \dots a_n$ in C_1^j takes at most $2(i - j)$ steps because each symbol x_k , $j + 1 \leq k \leq i$, located at its appropriate position, takes at most two steps. Node s' then connects to node $s'' = *^j x_{j+1} x_{j+2} \dots x_i \text{next}(a_{i+1}) a_{i+2} \dots a_n$ in C_1^j via at most two links. Therefore, the total number of steps for routing a message is at most $2(i - j) + 2$. $\square$

Based on the terms and basic lemmas presented above, two broadcasting algorithms will be addressed in the following section.

3. Broadcasting

In this section, we will introduce two optimal one-to-all broadcasting algorithms for incomplete star graphs. The first scheme is for single-message broadcasting and the second one is for multi-message broadcasting.

3.1. Single-message broadcasting

We first describe the basic idea of this scheme. Assume that the source node in class C^i needs to deliver a message to all the other nodes. The source node sends the message to one node of the first substar in C^i according to the labeling scheme. This node delivers the message to one node of the last substar in this class and to one node of the last substar in the previous class consecutively. To recursively deliver the source message in this manner, one node of each substar in all of the classes can receive the message from the source node. We know that a class is composed of substars with the same dimension. For each class, the node with the message transmits the mes-

sage to the other substars. Thus, one node of each substar in the incomplete star graph has received the message. By using the broadcasting algorithm in [13], we can independently perform the one-to-all broadcasting on each substar in parallel.

In the following lemma, we shall show that a certain number of steps are required for a node of the first substar in each class to receive a message from the source node.

Lemma 3. It takes at most $6n - 10$ steps for a source node to send a message to each relay node that is a node of the first substar of the corresponding class.

Proof. The worst case scenario is that there are several substars in each class and that the source node is in class C^{n-1} . This is because, if the source node is located in the middle of these classes, routing will be independent along the two opposite directions in parallel. Thus, we assume that substars in each of the classes $C^{n-1}, C^{n-2}, \dots$, and C^1 exist and the source node is in class C^{n-1} . The proof can be easily examined from Fig. 2. It takes two steps for the source node to deliver a message to node a in the first substar. By Lemma 1, node a sends the message to node b of the last substar. That takes two steps. By Lemma 2, node b sends the message to a node of the first substar in class C^{n-2} , which takes $2((n-1) - (n-2)) + 2 = 4$ steps. The two previous operations for sending a message to a node in C_1^1 takes $2 + \sum_{i=1}^{n-2} (2 + 4) = 6n - 10$ steps. $\square$

For example, consider broadcasting in a 17-node incomplete star graph, as shown in Fig. 3. Assume that the source node is 2143 in $***3$. First, node 2143 sends a message to node 3124 of the first substar C_1^3 in C^3 via node 4123. Node 3124 sends

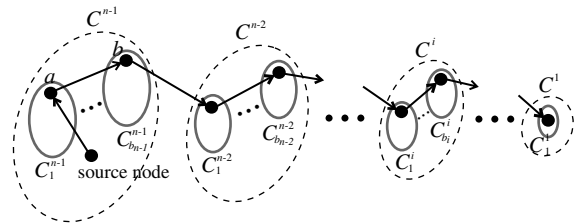


Fig. 2. Routing among classes.

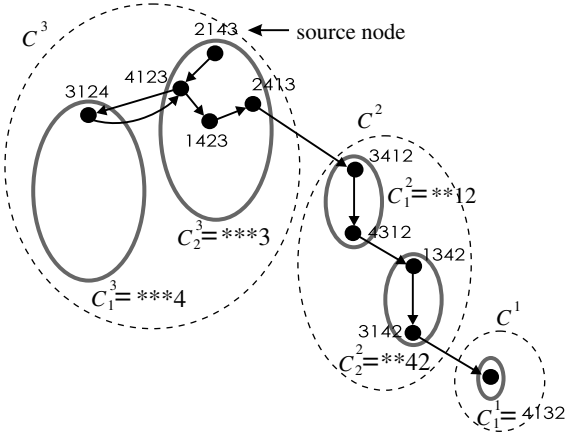


Fig. 3. Routing on a 17-node incomplete star graph.

the message to node 4123 of the last substar C_2^3 in C^3 . Node 4123 sends the message to node 2413 via node 1423. Node 2413 sends the message to node 3412 of the first substar C_1^2 in C^2 . Node 3412 sends the message to node 1342 of the last substar C_2^2 in C^2 via node 4312. Node 1342 delivers the message to node 4132 in C_1^1 via node 3142. Hence, nodes 3124, 3412, and 4132 on the first substars in classes C^3 , C^2 , and C^1 , respectively, receive the message from the source node.

The above-mentioned process routes a message among classes. In the following, we describe routing among substars in the same class. We shall show that a certain number of steps are required for a message to be delivered from a node to another node in each of the substars in the same class.

Lemma 4. *In class C^i , there are substars $C_1^i, C_2^i, \dots, C_{b_i}^i$. Assume that node a_1 in C_1^i has received a broadcasted message. It takes at most $2\lceil \log b_i \rceil$ steps for the message to be delivered from node a_1 to one node in each of the substars in C^i .*

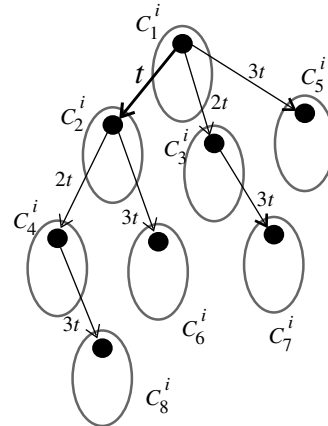
Proof. Node a_1 in C_1^i sends the message to node a_2 in C_2^i , which takes at most two steps. From Corollary 1, a_1 and a_2 send the message in parallel to nodes a_3 and a_4 in C_3^i and C_4^i , respectively. Finally, it takes at most $2\lceil \log b_i \rceil$ steps to broadcast the message to one node in each of the substars in the class. $\square$

For example, assume there are eight substars $C_1^i, \dots, C_8^i$ in class C^i . First, a node in C_1^i sends the message to a node in C_2^i . Then, the two nodes in C_1^i and C_2^i send the message in parallel to two nodes in C_3^i and C_4^i , respectively. Finally, four nodes in C_1^i, C_2^i, C_3^i , and C_4^i send the message in parallel to nodes in C_5^i, C_6^i, C_7^i , and C_8^i , respectively. The above-mentioned process is shown in Fig. 4. Note that nt in Fig. 4 denotes the n th step time in the routing.

Until now, one node within all of the substars making up the incomplete star graph receives the message from the source node. Next, we can perform broadcasting on these substars in parallel by using the proposed algorithm in [13] to accomplish this operation. For one-to-all broadcasting, the lower bound of time complexity is $O(\log N) = O(n \log n)$, where $(n-1)! < N < n!$.

Theorem 1. *The single-message broadcasting algorithm takes $O(n \log n)$ steps, which is optimal, for an N -node incomplete star graph.*

Proof. First, the source node sends a message to each relay node that is a node of the first substar of the corresponding class. From Lemma 3, it takes at most $O(6n-10) = O(6n)$ steps. Second, the node (relay node) delivers the message to one node in each of the substars in the class. From Lemma 4, it takes at most $O(\max_{1 \leq i \leq n-1} (2\lceil \log b_i \rceil)) = O(\log(n-1))$ steps. Thus, one node

Fig. 4. Routing in class C^i .

of each substar making up the N -node incomplete star graph has received the broadcasted message. Finally, based on the proposed broadcasting algorithm in [13], broadcasting in each substar takes $O((n-1)\log(n-1))$ steps. Thus, the time complexity of our proposed algorithm is $O(6n) + O(\log(n-1)) + O((n-1)\log(n-1)) = O(n\log n)$, which is optimal. $\square$

3.2. Multi-message broadcasting

Due to the need to transmit a large amount of data from the source node to all other nodes in some applications, we can have to divide the original data into several packets (messages) for broadcasting. In this subsection, an optimal algorithm is presented for multi-message broadcasting on incomplete star graphs. The basic idea for broadcasting is described as follows.

Assume that the source node in class C^i needs to deliver a stream of m messages to all the other nodes. First, we will build a path to connect each class. The source node can be connected with one node of the first substar in C^i . This node of the first substar can be connected to one node of the last substar in the class C^i and to one node of the last substar in the previous consecutive class C^{i-1} . We recursively construct a routing path so that the path connects the first substar with the last substar in each class. Thus, we can transmit these messages in a pipeline fashion via this established path. Using the same idea as this construction, we can build a path connecting each substar in all of the classes. In this way, we can also transmit the messages in a pipeline fashion via this constructed path. So far, one node of each substar in the incomplete star graph has received the m messages. Using the broadcasting algorithm in [14], we can independently perform the one-to-all broadcasting with multiple messages on each substar in parallel.

First, by Lemma 3, we can construct a path with length of at most $6n - 10$, which connects the nodes of the first substar in each class. In this constructed path, we can use the same links but use different directions. If the source node is in C^{n-1} or C^1 , the transmission of a stream of m mes-

sages in a pipeline fashion takes $O(n + m)$ steps. Otherwise, it takes at most two times $O(n + m)$ steps. Thus, this operation takes $O(n + m)$ steps.

Then, within class C^i , the transmission of m messages proceeding in a pipeline fashion, as shown in Fig. 5, takes $O(b_i + m)$ steps for $1 \leq i \leq n - 1$. Due to the operation performed in parallel for each class, it takes $O(\max_{1 \leq i \leq n-1} b_i + m) = O(n + m)$ steps for one node of each substar to receive m messages from the node of the first substar that has these messages.

Finally, we can apply the proposed algorithm in [14] to accomplish this broadcasting. In [14], the number of steps needed to broadcast m messages to all the other nodes in an n -star is $O(n\log n + m)$, which is the optimal number of steps. Thus, broadcasting m messages to all the other nodes in an N -node incomplete star graph in $O(n\log n + m)$ steps is optimal. The proposed broadcasting algorithm takes $O(n\log n + m)$ steps which will be proven below.

Theorem 2. *The multi-message broadcasting algorithm takes $O(n\log n + m)$ steps, which is optimal, for an N -node incomplete star graph.*

Proof. We know that the path we constructed has a length of at most $6n - 10$ which connects the nodes of the first substar in each class. Transmitting a stream of m messages in a pipeline fashion takes $O(n + m)$ steps. For each class, it takes

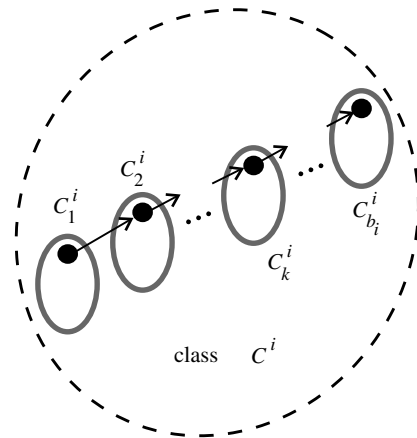


Fig. 5. Pipeline transmission in class C^i .

$O(n + m)$ steps for one node of each of the substars to receive m messages from a node of the first substar. One node of each complete substar will deliver these m messages to all the other nodes, which takes $O(\max_{1 \leq i \leq n-1}(i \log i + m)) = O((n-1) \log(n-1) + m)$ steps. Hence, the time complexity of this algorithm is $O(n + m) + O(n + m) + O((n-1) \log(n-1) + m) = O(n \log n + m)$, which is optimal. $\square$

4. Conclusions

In this paper, we proposed two optimal one-to-all broadcasting algorithms for incomplete star graphs with a single-port communication model. The first scheme takes $O(n \log n)$ steps for one-to-all broadcasting on an N -node incomplete star graph, where $(n-1)! < N < n!$. The second one takes $O(n \log n + m)$ steps for broadcasting m messages in a pipeline fashion on the same incomplete star graph.

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